

SPARSE SAMPLING AND BOREL REFINEMENTS OF CLOSED COVERS

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ABSTRACT. Let T be conjugate to a positive power of a mixing one-sided shift of finite type, and let $\mathcal{K}$ be a finite closed cover of its state space. We prove that, along every increasing sequence whose successive gaps tend to infinity, the minimum combinatorial sequence entropy over all finite Borel partitions refining $\mathcal{K}$ is $\log N(\mathcal{K})$, where $N(\mathcal{K})$ is the minimum size of a subcover. The same partition attains this value for all such sequences and minimizes maximal pattern entropy. The sampling sequence is fixed before the Borel partition is chosen. The proof uses the nonmeager atoms of a partition, openness of the shift, and finite-type concatenation. Applied to safe control blocks with a common successor, the result gives an exact fixed-period minimax formula over all legal Borel feedbacks, including arbitrary Borel mixtures and atom refinements. A six-control system with two possible successors at each state has minimum safe-cover size three but full-feedback minimax value $\log 2$. Thus the cover formula need not survive feedback-dependent successors, even with clopen safety domains and continuous open branches.

1. INTRODUCTION

For a finite partition observed along an orbit, one can count the names seen at a prescribed sparse set of times. Two choices then arise: the sampling sequence and the observation partition. When observations must refine a given cover, it is natural to ask whether a single sequence, chosen before the partition, detects the least possible complexity. For Borel partitions this is not a finite-window question: the atoms need not contain open sets, and the partition may depend on the entire state.

This paper determines that optimization for finite closed covers of mixing shifts of finite type. If r is the minimum size of a subcover, then every sequence with gaps tending to infinity gives a lower bound $\log r$ for *every* finite Borel refinement. An ordered difference partition of a minimum subcover attains the bound. In particular, the sequence $(0, 1, 4, 9, \dots)$ can be specified in advance. The theorem is about a common minimum threshold; it does not assert that this sequence maximizes the entropy of each individual partition.

The distinction between a cover and its refining partitions is essential. Romagnoli [7] computed, on the two-sided full shift over $q > 2$ symbols, a clopen cover with consecutive-time cover entropy $\log(q/(q-1))$ and minimum consecutive-time entropy $\log 2$ among its clopen refinements. The cover excludes the current symbol. Our result allows all Borel refinements, uses prescribed sparse times, and applies to arbitrary finite closed covers of mixing finite-type shifts. We spell out the corresponding sparse cover-partition gap in Corollary 3.1. This does not settle the consecutive-time Borel refinement question discussed in [7].

Common sampling and independence already have substantial precedents. Huang, Maass, and Ye [1, Theorem 2.3] identify the attained supremum of measure sequence entropies of a fixed partition. More particularly, their Lemma 3.3 and Theorem 3.5 give conditions on a measurable cover under which there is *one* sampling sequence with a strictly positive lower bound for the measure sequence entropies of *all* its finite measurable refinements. Hence a common positive threshold over Borel refinements, without further qualification, is not the

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contribution asserted here. Their bound uses an invariant measure and a sequence obtained from that cover; it is not the exact combinatorial value $\log N(\mathcal{K})$ for every prescribed sequence with diverging gaps. The measure-cover quantities and their relation to independence are developed further by Kerr and Li [5, Section 4 of the author version].

Li [6, Theorem 2] characterizes topological weak mixing by sequence-entropy attainment for each regular open cover, with the sequence chosen for that cover. Huang and Ye [2, Theorem 2.1] prove attainment of maximal pattern entropy by sequence entropy for a fixed open cover, as well as a measurable counterpart. Independence of finite patterns is part of the established framework of Kerr and Li [4]. We use the elementary finite-type form of this independence and give its proof. The additional step here is to pass from nonmeager Borel atoms to actual orbit patterns on every prescribed sequence with diverging gaps, and to convert closed-cover refinement into an exact cardinality optimization. Neither independence itself nor a new general variational principle is claimed.

There is also a control interpretation. A safe feedback assigns a finite control block to each state and repeats this assignment at a fixed period. If all legal blocks have the same endpoint map, feedback does not alter the state dynamics at block boundaries, but may choose among overlapping safety domains in any Borel fashion. Theorem 4.2 computes the full-feedback minimax value as $\log r/\tau$, without restricting feedbacks to continuous or finite-window rules. A positive example has overlapping safety domains and no nonconstant control-letter projection forced by every feedback. Finally, we give an exact example in which feedback changes the successor: the same cover number is three, while the full-class minimax value is $\log 2$. This is a boundary of the cover formula, not a counterexample to minimax equality itself.

2. COUNTING ENTROPY AND THE CLOSED-COVER THEOREM

Let Q be a nonempty compact metric space and let $T : Q \rightarrow Q$ be a Borel map. For a finite Borel partition α of Q and a finite $F \subset \mathbb{N}_0$, put

$$P_\alpha(F) = \left| \left\{ (\alpha(T^k x))_{k \in F} : x \in Q \right\} \right|,$$

where $\alpha(y)$ denotes the atom containing y and empty atoms are discarded. Thus $P_\alpha(F)$ is the number of nonempty atoms in $\bigvee_{k \in F} T^{-k} \alpha$. For a strictly increasing sequence $A = (a_i)_{i \geq 1}$ in $\mathbb{N}_0$, define

$$h_A(T, \alpha) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log P_\alpha(\{a_1, \dots, a_n\}), \quad (2.1)$$

$$p_\alpha^*(n) = \max_{\substack{F \subset \mathbb{N}_0 \\ |F|=n}} P_\alpha(F), \quad h^*(T, \alpha) = \lim_{n \rightarrow \infty} \frac{1}{n} \log p_\alpha^*(n). \quad (2.2)$$

All logarithms are natural. These are combinatorial counting quantities; for Borel atoms they are not Shannon entropies of an invariant measure. The maximum in (2.2) exists because the counts are integers between 1 and $|\alpha|^n$. Splitting a window into two parts gives $p_\alpha^*(n+m) \leq p_\alpha^*(n)p_\alpha^*(m)$. Subadditivity proves the existence of the limit, and $h_A(T, \alpha) \leq h^*(T, \alpha)$. No invertibility or continuity of the observation is used here.

Let $\mathcal{K} = (K_w)_{w \in W}$ be a finite closed cover of Q , and write

$$r = N(\mathcal{K}) = \min\{|I| : I \subset W, \bigcup_{w \in I} K_w = Q\}.$$

The set $\mathcal{B}(\mathcal{K})$ consists of all finite Borel partitions refining $\mathcal{K}$: each atom must be contained in at least one K_w . Equivalently, one may first consider all Borel maps $s : Q \rightarrow W$ with $x \in K_{s(x)}$, and then allow arbitrary finite Borel refinements of their fibers.

Let H be a primitive finite 0–1 matrix, with vertex set V , and let

$$Y_H = \{y \in V^{\mathbb{N}_0} : H_{y_j y_{j+1}} = 1 \text{ for all } j \geq 0\}.$$

Primitivity means that all sufficiently large powers of H have strictly positive entries. Write σ for the one-sided shift.

Theorem 2.1. *Suppose $T : Q \rightarrow Q$ is topologically conjugate to $\sigma^p : Y_H \rightarrow Y_H$ for some primitive H and integer $p \geq 1$. Let $\mathcal{K}$ be any finite closed cover of Q . For every strictly increasing $A = (a_i)$ satisfying*

$$a_{i+1} - a_i \rightarrow \infty, \quad (2.3)$$

one has

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \min_{\alpha \in \mathcal{B}(\mathcal{K})} h^*(T, \alpha) = \log r. \quad (2.4)$$

An ordered difference partition of any minimum subcover attains both minima, simultaneously for every A satisfying (2.3). Consequently,

$$\max_A \inf_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(T, \alpha) = \log r, \quad (2.5)$$

where the maximum ranges over all strictly increasing infinite sequences and is attained, for example, by $a_i = (i - 1)^2$.

The order of quantifiers is part of the assertion. Every A satisfying (2.3) works for every allowed partition, although the finite initial segment discarded in the proof can depend on that partition. The proof does not interchange an infimum over partitions with a finite-window limit.

2.1. A category and concatenation lemma. We use meagerness relative to Q . A Borel set has the Baire property, so a nonmeager Borel set is comeager in some nonempty open set. Indeed, the sets differing from open sets by meager sets form a sigma-algebra: for complements, the additional difference is contained in the nowhere dense boundary of an open set. A compact metric space is a Baire space. In particular, a nonempty open set cannot be covered by countably many nowhere dense sets, as is also seen by choosing nested nonempty open sets with closures avoiding those sets and diameters tending to zero.

If $F : Q \rightarrow Q$ is continuous and open, preimages of meager sets are meager. For a closed nowhere dense N , the closed set $F^{-1}N$ has empty interior: otherwise the image of a nonempty open subset would be a nonempty open subset of N . Passing to countable unions proves the assertion.

Lemma 2.2. *Under the dynamical assumptions of Theorem 2.1, T is continuous, open, and onto. Given finitely many nonempty cylinder sets $C_1, \dots, C_q$ in symbolic coordinates, there is an integer $b \geq 1$ such that, for every $k \geq 1$, every $t_1, \dots, t_k \in \{1, \dots, q\}$, and all $0 \leq n_1 < \dots < n_k$ with $n_{j+1} - n_j \geq b$,*

$$\bigcap_{j=1}^k T^{-n_j} C_{t_j}$$

is nonempty and open.

Proof. Work on Y_H through the conjugacy. The image under σ of a prefix cylinder of length at least two is the cylinder obtained by removing its first symbol. The image of a one-symbol cylinder is the union of the permitted successor cylinders. Hence σ is open. Every vertex has a predecessor, so every infinite path has a predecessor; thus σ is onto. These properties pass to powers and conjugates.

Choose within each C_j a nonempty prefix cylinder, and extend their defining words, if necessary, to a common length ℓ . Choose N with $H^n > 0$ for every $n \geq N$, and then choose b with $pb - \ell + 1 \geq N$. Place the word for C_{t_j} at coordinate pn_j . Between the end of one placed word and the start of the next there are $p(n_{j+1} - n_j) - \ell + 1$ edges to fill. Primitivity supplies a path of that length between any two endpoint vertices. Predecessors fill the finite segment before the first placed word, and successors provide an infinite continuation after

the last. This constructs an actual point in the intersection. The intersection is open because it is a finite intersection of preimages of open sets under continuous maps. $\square$

Lemma 2.3. *Under the same assumptions, let α be a finite Borel partition with q nonmeager atoms. For every A satisfying (2.3) there exists $m \geq 1$ such that*

$$P_\alpha(\{a_1, \dots, a_n\}) \geq q^{n-m+1} \quad (n \geq m). \quad (2.6)$$

In particular $h_A(T, \alpha) \geq \log q$.

Proof. List the nonmeager atoms as $D_1, \dots, D_q$. The Baire property provides a nonempty cylinder C_j for which $C_j \setminus D_j$ is meager. Let b be supplied by Lemma 2.2, and choose m so that $a_{i+1} - a_i \geq b$ for all $i \geq m$.

Fix $n \geq m$ and any target labels $t_m, \dots, t_n$. The set

$$G = \bigcap_{i=m}^n T^{-a_i} C_{t_i}$$

is nonempty and open. Each power T^{a_i} is continuous and open, so

$$\bigcup_{i=m}^n T^{-a_i} (C_{t_i} \setminus D_{t_i})$$

is meager and cannot cover G . A point outside this union realizes all the specified atoms at the sampled times. Distinct target label strings give distinct patterns; restriction from the full window to this tail is onto. This proves (2.6). Dividing its logarithm by n and taking a limit inferior proves the asserted bound. $\square$

2.2. Proof of the exact formula.

Proof of Theorem 2.1. Let $\alpha \in \mathcal{B}(\mathcal{K})$. Assign to each atom D one index $w(D)$ with $D \subset K_{w(D)}$. Let J be the set of indices assigned to the nonmeager atoms. If the closed sets $(K_w)_{w \in J}$ did not cover Q , their complement would be a nonempty open set covered by the finitely many meager atoms of α , which is impossible. Thus $|J| \geq r$, and the number q of nonmeager atoms is at least r . Lemma 2.3 gives

$$h_A(T, \alpha) \geq \log q \geq \log r$$

for every A satisfying (2.3). Closedness of the cover members is used precisely in this passage from category to a full cover.

Choose a minimum subcover $K_{w_1}, \dots, K_{w_r}$, and set

$$E_j = K_{w_j} \setminus \bigcup_{k < j} K_{w_k}, \quad \alpha_0 = \{E_1, \dots, E_r\}. \quad (2.7)$$

Every E_j is Borel and is nonempty, since a minimum subcover has no redundant member. The partition refines $\mathcal{K}$ and has r atoms. Hence $P_{\alpha_0}(F) \leq r^{|F|}$ for all finite F , giving $h_A(T, \alpha_0) \leq h^*(T, \alpha_0) \leq \log r$ for every A . The established lower bound and $h^* \geq h_A$ yield (2.4) and simultaneous attainment. For arbitrary A , the partition α_0 bounds the inner infimum in (2.5) by $\log r$. Any A satisfying (2.3) gives the reverse bound and attains the maximum. $\square$

3. A COVER-PARTITION GAP AT PRESCRIBED SPARSE TIMES

For any finite cover $\mathcal{K}$ define its combinatorial sequence entropy by

$$h_A(T, \mathcal{K}) = \limsup_{n \rightarrow \infty} \frac{1}{n} \log N \left(\bigvee_{i=1}^n T^{-a_i} \mathcal{K} \right).$$

For a partition this is (2.1). For overlapping sets, covering each finite orbit window need not correspond to one fixed refining partition used at every time.

Corollary 3.1. *Let $q \geq 3$, $Q = \{0, \dots, q-1\}^{\mathbb{N}_0}$, $T = \sigma$, and*

$$K_j = \{x : x_0 \neq j\} \quad (0 \leq j < q), \quad \mathcal{K} = (K_j)_{j=0}^{q-1}.$$

For every increasing infinite A ,

$$h_A(\sigma, \mathcal{K}) = \log \frac{q}{q-1}. \quad (3.1)$$

For every such A with diverging successive gaps,

$$\min_{\alpha \in \mathcal{B}(\mathcal{K})} h_A(\sigma, \alpha) = \log 2 > h_A(\sigma, \mathcal{K}). \quad (3.2)$$

Proof. The cover has no single member equal to Q , and any two distinct members cover Q . Thus $r = 2$, and (3.2) follows from Theorem 2.1 once (3.1) is proved.

For a window F of size n , the full shift has all q^n possible input vectors. Each member of the joined cover specifies one excluded letter at each coordinate, and covers $(q-1)^n$ of these vectors. Consequently the minimum number $c(n)$ of such boxes is at least $(q/(q-1))^n$. For completeness, the following elementary probabilistic covering bound gives the matching exponent. Choose R excluded-letter vectors independently and uniformly from the q^n possibilities. A fixed input vector is left uncovered with probability at most

$$\left(1 - \left(\frac{q-1}{q}\right)^n\right)^R \leq \exp\left(-R \left(\frac{q-1}{q}\right)^n\right).$$

Taking $R = \lceil (q/(q-1))^n (n \log q + 1) \rceil$ makes the sum of these probabilities over all input vectors smaller than one. Some choice therefore covers every vector. We have proved

$$\left(\frac{q}{q-1}\right)^n \leq c(n) \leq \left\lceil \left(\frac{q}{q-1}\right)^n (n \log q + 1) \right\rceil.$$

The count depends only on n , not on the positions of the sampled coordinates, so its exponential rate proves (3.1). $\square$

The excluded-letter cover and its covering exponent are already present in Romagnoli [7]; the calculation above is included to identify the sparse-time quantity without an external dependency. The consequence of Theorem 2.1 is the exact minimum over *all Borel* refinements for a sequence fixed in advance. It does not assert that every Borel refinement has consecutive-time entropy at least $\log 2$, or that every sampling sequence gives that lower bound.

4. FIXED-PERIOD SAFE FEEDBACKS WITH A COMMON SUCCESSOR

Let X be a compact metric space, U a nonempty finite control set, and $f_u : X \rightarrow X$ continuous for every $u \in U$. Let $Q \subset X$ be a nonempty compact controlled-invariant set: at each $x \in Q$ at least one control keeps the next state in Q . All infinite control strings are allowed. Fix an integer period $\tau \geq 1$ once and for all, and put $W = U^\tau$. For $w = (u_0, \dots, u_{\tau-1})$ write

$$\varphi(0, x, w) = x, \quad \varphi(j, x, w) = f_{u_{j-1}} \circ \dots \circ f_{u_0}(x),$$

and define

$$K_w = \{x \in Q : \varphi(j, x, w) \in Q \text{ } (0 \leq j \leq \tau)\}, \quad g_w(x) = \varphi(\tau, x, w). \quad (4.1)$$

The K_w are closed and cover Q . The coverage follows by choosing legal single-step controls successively for the fixed finite horizon.

The class $\mathcal{S}_\tau$ consists of *all* Borel selectors $s : Q \rightarrow W$ such that $x \in K_{s(x)}$. Its closed-loop block map is $T_s(x) = g_{s(x)}(x)$. All Borel mixtures of legal selectors belong to $\mathcal{S}_\tau$. No continuity or dependence on finitely many coordinates is imposed. A selector is used without an additional clock, register, or feedback memory.

For a finite window F , let

$$\mathbf{P}_s(F) = \left| \{(s(T_s^k x))_{k \in F} : x \in Q\} \right|.$$

Define $h_A(s)$ and $h^*(s)$ from these counts as in (2.1)–(2.2), but dividing by $n\tau$ rather than n . Thus normalization is per sampled control block, divided by its fixed length; it is *not* division by elapsed time τa_n . Put

$$L_\tau = \sup_A \inf_{s \in \mathcal{S}_\tau} h_A(s), \quad M_\tau = \inf_{s \in \mathcal{S}_\tau} h^*(s). \quad (4.2)$$

Always $L_\tau \leq M_\tau$.

The counts use actual closed-loop trajectories. If instead one takes the closure of their one-sided name set, the finite patterns are unchanged: a cylinder specifying finitely many coordinates is open, so if it meets that closure, it meets the original name set. Taking a name closure therefore introduces no extra finite-window witnesses.

Lemma 4.1. *The infima in (4.2), and the inner infimum at any fixed A , are unchanged if one allows every finite Borel safe partition with a control block assigned to each atom, and counts atom names rather than control-block names.*

Proof. Merge all atoms assigned the same block. Their finite union is Borel and remains inside that block's safety domain. The resulting selector has exactly the same closed-loop state map. The map from original atom names to block names is coordinatewise and onto the actual block-name patterns, so every block-pattern count is no larger than the original atom-pattern count. Conversely, the nonempty fibers of any legal selector form a legal finite Borel safe partition whose atom and block counts agree. Both directions of the infimum equality follow. $\square$

Theorem 4.2. *Assume in addition that there is a map $T : Q \rightarrow Q$ such that*

$$g_w(x) = T(x) \quad \text{for every } w \in W \text{ and every } x \in K_w, \quad (4.3)$$

and that T is conjugate to σ^P on a mixing one-sided shift of finite type. With $r = N((K_w)_{w \in W})$, for every A satisfying (2.3) one has

$$\min_{s \in \mathcal{S}_\tau} h_A(s) = \min_{s \in \mathcal{S}_\tau} h^*(s) = L_\tau = M_\tau = \frac{\log r}{\tau}. \quad (4.4)$$

The maximum defining L_τ is attained by every such A , and an ordered minimum-subcover selector attains all the indicated minima. The same assertions hold over all finite Borel safe partitions and their arbitrary finite Borel refinements.

Proof. For every legal selector, (4.3) gives $T_s = T$ pointwise. Its fibers form a finite Borel refinement of $(K_w)_{w \in W}$. Theorem 2.1, with normalization by τ , gives the lower bound at every A with diverging gaps. If $w_1, \dots, w_r$ form a minimum subcover, define $s_0(x) = w_j$ on the sets E_j of (2.7). This is Borel and legal at every point. Each selected block stays in Q at its intermediate times by (4.1), and the endpoint again lies in Q , so recursion gives an infinite safe closed-loop orbit for every initial state. The selector uses only r blocks and thus gives the upper bound for every window and every sampling sequence. This proves attainment, (4.4), and the outer maximum. Lemma 4.1 proves the last assertion. $\square$

The standard safe-partition framework is described by Kawan [3, Chapter 2]. His minimization over control periods in invariance entropy is a different optimization. Here τ is fixed, intermediate safety is retained, and the exact value comes from the additional common-successor and finite-type hypotheses.

4.1. A positive overlapping example at period two. Take

$$Q = \{0, 1, 2\}^{\mathbb{N}_0}, \quad X = Q \sqcup \{\dagger\}, \quad U = \{0, 1, 2\}, \quad \tau = 2,$$

where the isolated point $\dagger$ lies outside Q . Define

$$f_u(x) = \begin{cases} \sigma x, & x \in Q, \ x_0 \neq u, \\ \dagger, & x \in Q, \ x_0 = u, \end{cases} \quad f_u(\dagger) = \dagger.$$

Each f_u is continuous since the cases are clopen. For $w = (a, b)$,

$$K_{(a,b)} = \{x : x_0 \neq a, x_1 \neq b\}, \quad g_{(a,b)}|_{K_{(a,b)}} = \sigma^2.$$

All nine control blocks are available to Borel feedbacks, and each state permits four of them. A safety domain covers four of the nine two-symbol input blocks, so two domains cannot cover Q . The three diagonal blocks $(0, 0), (1, 1), (2, 2)$ do cover Q , since some symbol differs from both x_0 and x_1 . Hence $r = 3$, and

$$L_2 = M_2 = \frac{\log 3}{2}.$$

An attaining selector is

$$c(x) = \min(\{0, 1, 2\} \setminus \{x_0, x_1\}), \quad s_0(x) = (c(x), c(x)).$$

It is clopen and legal everywhere. Its actual block-boundary states are $\sigma^{2k}x$. To prescribe any sequence of diagonal labels, use successive input blocks 11, 00, and 01 for the desired labels 0, 1, and 2, respectively. Therefore its actual name set is the full shift on these three blocks. The lower bound, however, holds for every legal Borel rule using any of the nine blocks.

This example does not preassign a nonconstant common control-letter observation to every feedback. More precisely, suppose a map $\rho : U^2 \rightarrow V'$ and an observation $F : Q \rightarrow V'$ satisfy $\rho(s(x)) = F(x)$ for every legal Borel selector and every x . Any two blocks (a, b) and (a', b') are colegal at a state beginning with ij , where $i \notin \{a, a'\}$ and $j \notin \{b, b'\}$. A legal baseline selector may be changed at this single state to use either block, while remaining Borel and legal. Hence their ρ values agree. All pairs of blocks are colegal somewhere, so ρ is constant. This rules out this particular forced-letter explanation; it makes no claim about every possible dynamical factor.

5. A FEEDBACK-DEPENDENT BOUNDARY WITH AN EXACT FULL-CLASS VALUE

The common-successor assumption cannot be removed solely because each branch is continuous and open. We give a system with individual prefix cylinders as safety domains and compute its full-class value.

Let $Q = \{0, 1\}^{\mathbb{N}_0}$, let $\mathcal{C} = \{0, 10, 11\}$ be a set of finite words, and take $U = \mathcal{C} \times \{0, 1\}$ and $\tau = 1$. For $e \in \{0, 1\}$ let P_e add e modulo two to every coordinate. On $X = Q \sqcup \{\dagger\}$ define

$$f_{(c,e)}(x) = \begin{cases} P_e \sigma x, & x \in [c], \\ \dagger, & x \in Q \setminus [c], \end{cases} \quad f_{(c,e)}(\dagger) = \dagger. \quad (5.1)$$

The cylinders $[0], [10], [11]$ are disjoint and cover Q . Each state has exactly two legal controls with different successors. The global maps $f_{(c,e)}$ are continuous, and their legal restrictions are open. The safety domains are $K_{(c,e)} = [c]$, and $r = 3$.

Proposition 5.1. *For the system (5.1), over all legal Borel feedbacks and arbitrary Borel mixtures,*

$$\min_s h_{A_0}(s) = \min_s h^*(s) = \max_A \inf_s h_A(s) = \log 2 < \log r, \quad A_0 = (0, 1, 2, \dots). \quad (5.2)$$

There is one clopen legal feedback s_f satisfying $h_A(s_f) = \log 2$ for every increasing infinite A . The same full-class values hold if arbitrary finite Borel atom refinements are allowed.

Proof. Let $c(x)$ be the unique member of $\mathcal{C}$ with $x \in [c(x)]$, and set

$$s_f(x) = (c(x), x_1), \quad T_f x = P_{x_1} \sigma x.$$

This is a clopen selector, legal at every state. Its four nonempty initial control fibers, for labels $(0, 0), (0, 1), (10, 0), (11, 1)$, are $[00], [01], [10], [11]$, respectively. For all $n \geq 1$, direct induction gives

$$T_f^n x = P_{x_n} \sigma^n x, \quad s_f(T_f^n x) = (0, x_{n+1} \oplus x_n), \quad (5.3)$$

where $\oplus$ is addition modulo two. In the induction step the chosen flip is $x_{n+1} \oplus x_n$; it cancels the previous flip and leaves $P_{x_{n+1}}\sigma^{n+1}x$. The first coordinate is always zero after time zero, which also proves all-time safety.

For every finite nonempty window F , with $n = |F|$, one has

$$P_{s_f}(F) = \begin{cases} 2^n, & 0 \notin F, \\ 2^{n+1}, & 0 \in F. \end{cases} \quad (5.4)$$

The four initial labels and two later labels give the upper bounds. To prove the lower bounds, prescribe any of the four initial labels if $0 \in F$, thereby fixing (x_0, x_1) ; if $0 \notin F$, set $x_0 = x_1 = 0$. At positive sampled times prescribe the desired second components d_k , and put $d_k = 0$ at the other positive times. Recursively set $x_{k+1} = x_k \oplus d_k$ for every $k \geq 1$. This constructs a full state in Q whose actual trajectory under the same selector has the required sampled controls by (5.3). Thus all counted patterns are realized by true infinite closed-loop trajectories. Equation (5.4) implies $h_A(s_f) = h^*(s_f) = \log 2$ for every A .

For the opposite bound fix any legal Borel selector s . If its first n actual controls from x are (c_k, e_k) , let $E_0 = 0$ and $E_k = e_0 \oplus \cdots \oplus e_{k-1}$. Since shifts commute with flips,

$$T_s^k x = P_{E_k} \sigma^k x.$$

Let $b(c)$ be the first bit of the word c . Safety at time k gives

$$x_k = b(c_k) \oplus E_k \quad (0 \leq k < n). \quad (5.5)$$

Thus one actual length- n control word determines at most one input prefix of length n . Every binary input prefix has an extension in Q , and the feedback is defined and safe on every such extension. Distinct prefixes must yield distinct actual control words. Hence

$$P_s(\{0, \dots, n-1\}) \geq 2^n$$

for every n and every legal s . This proves $h_{A_0}(s) \geq \log 2$ and $h^*(s) \geq \log 2$. Combining these inequalities with s_f proves (5.2), including attainment. Lemma 4.1 handles atom refinements. $\square$

The obstruction has a precise orbit interpretation. In this example $T_f(Q) = [0]$: for the reverse inclusion, given $y \in [0]$, take $x = 0y$. Therefore neither $[10]$ nor $[11]$ is visited at a positive time. The initial fibers with those labels are nonmeager, but their labels cannot be freely concatenated in the future. The map T_f is itself continuous and open, as follows by restricting to the four clopen two-symbol cylinders. It is not onto Q .

Even preservation of meager preimages does not repair this problem. For any selector s and any n , the value of $T_s^n x$ is either $\sigma^n x$ or $P_1 \sigma^n x$. Hence, for a meager set N ,

$$(T_s^n)^{-1}N \subset (\sigma^n)^{-1}N \cup (P_1 \sigma^n)^{-1}N$$

is meager. What fails is the nonempty open intersection used in Lemma 2.2 for the *selected* dynamics. Also, the same s_f gives

$$\inf_s h_A(s) \leq \log 2 = \log r - \log(3/2) \quad \text{for every } A.$$

The failure of the cover formula is therefore a uniform numerical gap over all samplings, rather than a failure at one finite window. It does not refute $L_\tau = M_\tau$: equality holds in this very example. Nor does the proof determine the inner infimum for every other fixed sampling sequence.

6. SCOPE OF THE CONCLUSIONS

The exact closed-cover formula combines three specific features: closed safety constraints force enough nonmeager labels, an open common map preserves category under preimages, and finite-type concatenation realizes every finite choice of those labels at sufficiently separated times. This makes a prescribed divergent-gap sequence work for an uncountable family of

Borel observations without a countable approximation argument. The ordered subcover gives an actual minimizer, so no assumption about attainment of an abstract infimum is needed.

The control application retains every legal Borel selector, but has a substantial dynamical restriction: legal blocks share their endpoint. The six-control example identifies a failure of extending the cover number formula to changing successors. Its exact value uses an explicit normalizing feedback and input-prefix decoding; it does not supply a theory for arbitrary feedback-dependent branches. The equality $L_\tau = M_\tau$ for general finite-control, fixed-period systems over the full Borel class remains undecided by these results. No conclusion about simultaneous attainment of each feedback's individual maximal pattern entropy follows from the common minimum threshold proved here.

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